

Flexibility with channel count

Increased performance and lowering of ADC prices have also allowed the individual channels on a DAQ system to have their own ADC, rather than sharing a single ADC amongst different multiplexers, thereby making DAQ systems more granular in their channel count.

Such modules are designed to work with a specific signal type and provide a dedicated ADC on each channel, this facilitates lower channel count modules and devices, which you can group together to develop a custom DAQ system with the exact channel count and mix to fit a specific application need.

Larger share for PCIe, AXI and even USB buses

Although the commonly used platform for modular systems is PXI, but it is not the only modular platform solution available in the market. Other platforms, such as AXI, give you a larger port size, which is about four times of PXI. This allows you to put much more complex hardware and circuitry on the AXI platform.

In general, with ADC technology we have trade-offs with the sampling rate and the resolution. If we increase the sampling rate, resolution has to go down and if we increase the resolution, the sampling rate has to go down.

But with AXI, we can get ADCs which go up to 3.2GHz sampling rate and still maintain a good 12-bit resolution. This has been an achievement as there are very few digitisers crossing 1GHz sampling rate at 12-bit resolution.

USB is a very interesting product for DAQ systems. Providing added functionality for DAQ, compared to plug-in PCI bus devices, USB offers additional benefits such as auto-discovery, set-up and the ability to locate the device up to 5m from the PC with a standard USB cable. It improves both ease of use and flexibility. The bus apportioning up to 60MB/s of bandwidth, and signal streaming technologies have ensured the simultaneous operation of multiple DAQ data streams.

Vendors are actually aiming at enhancing user interface

Application software is as important as hardware in any DAQ system. To take full advantage of your hardware, your software should be simple enough to use and at the same time configurable enough, as just simplicity does not solve the problem. With the popularity of proprietary software on a decline, we have a wide range of software that are configurable in nature. You can make certain visual changes and get measurements done, while at the same time you are not required to have any programming knowledge.

These software have contributed to good user experience as well as utilising the hardware to its full capability. Easy and powerful DAQ software with high-performance, multi-threaded applications are available to provide a consistent set of functions across multiple programming languages such as C/C++, .NET and VB and different operating systems such as Windows, Linux and RTOS, for all supported PC buses.

Also, there are centralised GUIs which allow you to quickly visualise multiple measurements simultaneously and add multiple instruments to them, have multiple measurements, visualise and log the data if you want. Also, you can quickly export data to various applications and do not need any programming. Instead of having individual tools or software panels for each and every instrument, these tools unify all of these in a single software, allowing you to connect different instruments.

Most of the USB DAQ systems are standalone cards, which can directly work with a PC without a mainframe. Interestingly, the same card can be put into a chassis and then used as a modular system, where you can club different kinds of cards together or club multiple cards of the same type together to increase the channel density. You can use it as a standalone card if you are looking for only a few channels, or if you are looking for a higher channel density, the same card can be inserted into the chassis and then it can be controlled via a host PC.

Wireless approach is growing

While handheld computing platforms with USB interfaces are popular, newer versions having wireless communication have been developed. Designed for remote monitoring, operation and control of the system, these mobile computing devices are proving to be game changers for wireless DAQ systems.

Wide adoption of high-speed continuous transmission standard IEEE 802.11 (Wi-Fi) in mobile computing applications has forced DAQ system vendors to improve the industrial ruggedness of Wi-Fi products. The economies of scale, reliability and ease of use of Wi-Fi make the technology appropriate for integration with DAQ systems.

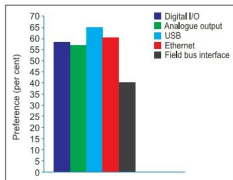


Fig. 2: Most preferred DAS system interface (Source: Integrated Process Systems)

Additionally, significant enhancements in the security and reliability of Wi-Fi networks, with technologies such as WPA2 data encryption, have contributed towards Wi-Fi being seen as the next promising technology for DAQ. Also, the seamless connection of Wi-Fi access points to Ethernet-based DAQ systems is adding to their utility.

The latest wireless DAQ devices come with built-in signal conditioning and high levels of network security, allowing you to stream data in real time for easy-to-use, high-performance remote wireless sensor measurements.

Also, there are different kinds of batteries available for the Wi-Fi DAQ devices with single use and rechargeable options. While the modern devices are designed to last a full day on a small to moderately sized battery, for longer applications batteries with recharging options (such as with a solar panel) are also available.